

Greater than the Sum of its Parts: Exploring Social Influence of Multi-Agents

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Abstract

Multi-agent systems, with multiple independent AI agents working together to achieve a common goal, are becoming increasingly prevalent in daily life. Drawing inspiration from the phenomenon of human group social influence, we investigate whether a group of AI agents can create social influence on users to agree with them, potentially changing their opinions. We conducted a study in which participants discussed their views toward two paintings with either a single or multiple AI agents, and where the agents either expressed like or dislike towards each painting. We found that conversing with multiple agents caused a greater shift in opinion towards the agents' stances on the paintings. Our study shows the potential advantages of multi-agent systems over single-agent ones in effecting opinion change. We discussed the resulting possibilities for multi-agent systems that promote social good, as well as for potential malicious actors to use these systems to manipulate public opinion.

CCS Concepts

• **Applied computing** → Psychology; • **Human-centered computing** → User studies.

Keywords

Social Influence, Multi-agent, LLM Agent

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1 Introduction

AI agents are advanced systems capable of interacting with users, holding conversations, and expressing their own thoughts [48]. These agents leverage the conversational and reasoning capabilities of large language models (LLMs) like Gemini, GPT-4, and DeepSeek, and have become commonplace in daily life [33]. The prevalence and ubiquity of LLM-powered agents has given rise to the concept of multi-agent systems—systems with two or more independent agents working together to fulfill a common goal and improve task performance [7, 16]. These systems represent an evolution in HCI research, offering solutions to complex challenges such as improving decision-making processes [28] and addressing filter bubbles [51]. Additionally, major AI companies such as Anthropic [11] and Google [15] have introduced multi-agent frameworks, making it easier for developers to create such interfaces.

With the rise of multi-agent systems in our daily lives, an interesting question emerges: *Can a group of AI agents collectively exert social influence on users?* This question is grounded in a well-established phenomenon in human-human interaction: individuals are more likely to conform to the opinions of larger groups, a concept central to social influence theory [9]. Social influence is particularly evident in contexts where opinions and preferences are shaped by collective input, such as product reviews [36, 44], public voting [5], or social media trends [31], where research has consistently shown that people's perceptions are influenced by the opinions of others in these scenarios. The theory further asserts that as group size increases, so does the pressure to conform. It is well-known that human-computer interaction dynamics tend to mirror human-human interaction dynamics [26], with humans perceiving and reacting to computer systems similar to how they would behave towards other humans. This similarity raises an important question: *Can a group of AI agents expressing the same opinion exert stronger influence on users than a single agent*, similar to the influence dynamics observed between groups and individuals in human interactions?

To investigate these questions, we conducted a preliminary study ($n = 50$) where participants discussed their stances on artworks with either *one* or *five* agents. Agent(s) would first express their like for one artwork, and then their dislike towards a second one. The

discussion content was kept consistent across conditions, differing only in the number of agent(s) presenting it. Survey data and open-ended responses about participants' perceptions towards the paintings were collected before and after each conversation round.

We found that the **multi-agent setup had significantly greater influence on participants' perceptions of the paintings**, as compared to the single-agent setup. Specifically, when the agent(s) said they liked the painting, participants in the 5-agent condition rated it more positively than those in the 1-agent condition; conversely, when the agent(s) expressed dislike towards the painting, participants in the 5-agent condition rated it more negatively than those in the 1-agent condition. These findings suggest that the number of virtual agents expressing a given opinion can affect the stance of human users. We discuss possible reasons for this phenomenon and relate it to analogous effects with human social groups.

2 Related Work

2.1 Social Influence Theory

When multiple individuals express the same opinion, others in the group often feel pressure to conform, a phenomenon known as "social influence" [9]. This concept, fundamental to social psychology, explains how people adapt their behavior to align with social expectations and has been extensively applied in various contexts, including marketing [36, 44, 52], health interventions [39, 50], sustainability [2, 46], and political discussions [32]. For example, Salganik et al. [36] demonstrated this effect in cultural markets, showing that participants were more likely to choose and listen to songs previously favored by others. This inspired the cultural context of our experiment setup, where we designed experiments to manipulate and evaluate people's attitudes towards the artworks (see Section 3.1 for details on study design rationale).

2.2 Multi-Agent Systems

AI agents are increasingly integrated into daily life, leveraging advanced conversational and reasoning capabilities for problem-solving tools [34, 49], educational support [22], and novel interfaces [24]. However, individual agents often lack domain expertise or robust reasoning for complex tasks [14]. To address this, studies have explored multi-agent techniques to improve the reasoning performance of language models [6, 13]. Multi-agent systems have enhanced natural language processing, software engineering, and robotics by simulating group dynamics [8, 16, 18]. They also simulate human roles, enabling collaboration and task sharing [23, 29]. Efforts in user-centered contexts have begun to explore multi-agent interactions, such as interfaces combining smart assistants [10] and frameworks enabling developers to deploy multi-agent systems [12, 15].

2.3 Multi-Agent and Human Interaction

As human-multi-agent collaboration becomes increasingly prevalent in daily life, the social influence exerted by multiple agents on humans in these interactions remains underexplored in the HCI domain. Understanding this dynamic is critical, as single AI agents have been shown to act as social actors, influencing opinions and behaviors [1, 3, 27, 38, 45]. Drawing from social influence theory in human society, it is plausible that groups of AI agents could exert

even stronger influence [19, 25], potentially leading to polarized opinions or manipulation. While previous research in HCI has primarily focused on designing multi-agents as tools—demonstrating their effectiveness in reducing cognitive load during information elicitation or supporting decision-making by providing diverse perspectives [28, 42]—their potential role as social actors capable of shaping user opinions and behaviors remains largely unexplored.

To address this research gap, we selected artworks as the experimental setting (see Section 3.1). Within this context, we formulated the following research questions:

RQ1: Could multi-agent convey stronger influence on people's perceptions towards artworks than single-agent?

RQ2: How, if any, does the design of single-agent and multi-agents influence people's attitudes towards the artworks?

3 Methods

To understand the different social influence effects of single- and multi-agent systems, we conducted a mixed-methods study combining an experiment and a survey. Within the survey, quantitative measures were used to address RQ1, while qualitative open-ended questions provided insights for RQ2.

3.1 Task Design

In our study, participants engaged with the agent(s) to discuss their perceptions of two paintings. The choice of paintings as a study topic was inspired by prior research on social influence in cultural markets [36]. Paintings were selected as they represent a relatable, everyday subject, allowing most individuals to express their opinions. The artworks were sourced from an open-access project on artistic preferences¹. A pilot study was conducted with members of our research group to assess people's initial attitudes toward the paintings. Based on the pilot results, we selected two paintings that received moderate ratings (neither strongly liked nor disliked). This selection aimed to minimize extreme preconceived opinions among participants, following criteria of selecting the experimental targets established in prior studies [43].

3.2 Experiment Setup

The experiment procedure was illustrated in Figure 1. Participants first completed a pre-survey that assessed their initial opinions on the paintings. They were then randomly assigned to one of two conditions: a single-agent (1-agent) or multi-agent (5-agent) setting. The choice of five agents for the multi-agent condition was based on prior research on multi-agent interface design, where five is considered a practical and manageable number for real-world social groups and applications [4, 21, 28].

In both conditions, participants were explicitly informed that they would interact with *AI agents* (Figure 2, left). A host agent would first appear to guide the procedure. The host would introduce the task and ask all agents and the participant to give a brief self-introduction. Then, participants engaged in two rounds of conversation with the assigned agent(s). In each conversation round, the host agent would first ask the agent(s) to share their opinions (like/dislike) on the painting, and then asked participants to express their feelings. After the participants shared their feelings, the

¹<https://openpsychometrics.org/tests/APS/>



Figure 1: Study procedure: Participants first completed a pre-survey to rate their initial attitudes toward the paintings. They were then assigned to one of two conditions (1-agent or 5-agent) and engaged in two rounds of conversations with the agent(s) about two paintings. Following the discussions, participants rated their final attitudes toward the paintings in the post-survey.

agent(s) would respond to the participants' statements of the paintings. Once all conversations were finished, participants completed a post-survey to capture their final attitudes on the two paintings.

The study was conducted in a self-developed online platform (as shown in Figure 2, right), and participants were required to complete it on a computer. The platform's frontend interface was built using JavaScript and HTML and included two main sections: (1) a login page and (2) a conversation page for interaction with agents.

3.3 Agent Setup

The agents' conversations were implemented through a combination of rule-based scripts and the GPT-4 API². For rule-based scripts, we designed a series of arguments either expressing like or dislike for the two paintings with arguments focused on art. These arguments were then crafted into agent dialogue, such as "I really like this one! The brushwork seems really delicate and the color choices are well-thought-out, so there's a sense of harmony." or "I don't think something like this is too suitable as a gift. My reason is simple - This painting depicts a person? It's a little strange to have that hanging in your house, don't you think?" Across the two conditions, the same set of arguments was presented: with five agents, the different agents took turns to present different arguments; with one agent, the same agent presented all the arguments in the same order (example dialog is shown in Figure 3 (right), and detailed scripts are presented in Appendix A.3.3). This was to ensure that if the two conditions led to different shifts in opinions, it was not because the argument quantity or quality presented in each condition varied.

We also integrated GPT-4 to enhance agent conversations by generating interactive responses during discussions. When the host agent prompted users to share their opinions on the paintings, the agent(s) would provide brief feedback based on user inputs, such as giving a summary or asking a question based on the stage of conversation. For example, if the participant said that "I think this painting looks pretty good", the next agent might have said "I'm

glad you like it! It's always interesting to hear what aspects of art resonate with different people." These responses were tightly regulated through detailed prompts (see Appendix A.3.3 for details), ensuring that the agent(s) only expressed understanding in concise messages, thereby preventing any issues related to AI hallucinations.

The agents' avatars and rhetorical styles were designed to appear human-like to enhance user acceptance [37]. To avoid the uncanny valley effect [40], we used cartoon-style avatars instead of realistic photos. Each avatar was themed around a unique set of colors to help participants distinguish between the agents. We also ensured gender balance within the 5-agent conditions to minimize the potential effect of agent gender on participants' opinion change [43].

3.4 Participants

Participants were recruited via CloudResearch³. The selection criteria required them to be English speakers and over 18 years old. A total of 50 participants were recruited and included in the analysis: 25 participants (F: 17, M: 8) in the 1-agent group, and 25 participants (F: 11, M: 14) in the 5-agent group. The average ages for each group were as follows: 1-agent group = 36.28 (SD = 11.28), and 5-agent group = 36.04 (SD = 10.66). Participants' educational backgrounds were as follows: 3 were high school graduates, 10 had some college but no degree, 5 held an associate's degree, 19 held a bachelor's degree, 12 held a master's degree or higher, and 1 preferred not to specify. The study lasted approximately 35 minutes to complete, and each participant was reimbursed US\$4.50.

3.5 Measurements

3.5.1 Quantitative Measures (RQ1). We adapted three items from the Art Reception Survey [17], a widely used tool for evaluating aesthetic perception, to assess participants' attitudes toward the paintings: artistic quality, positive attraction, and cognitive stimulation. For instance, artistic quality included questions such as "The composition of this painting is of high quality," cognitive stimulation included questions like "It is exciting to think about this painting,"

²gpt-4-1106-preview; <https://platform.openai.com/docs/models/gpt-4-turbo-and-gpt-4>

³<https://www.cloudresearch.com/>

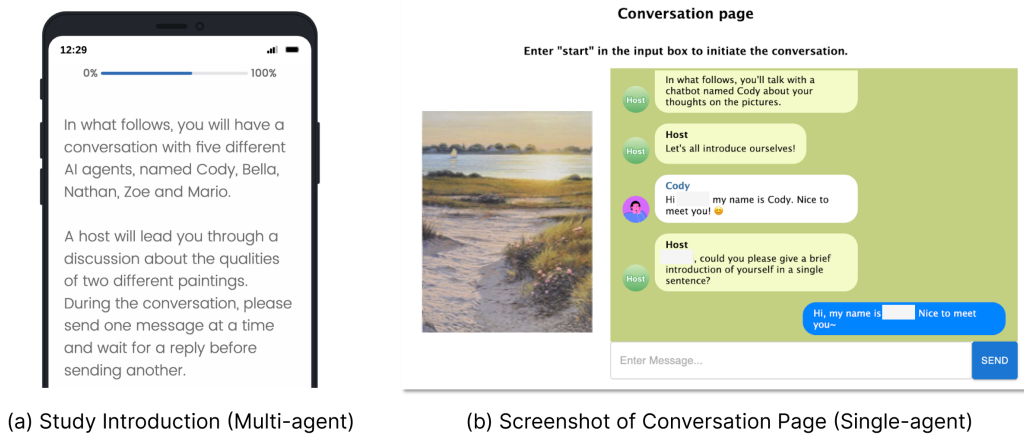


Figure 2: Screenshot of the study introduction and user interface. (a) The survey introduction, explicitly informing participants that they will interact with AI agent(s). (b) The user interface, displaying the painting on the left and the conversation interface on the right.

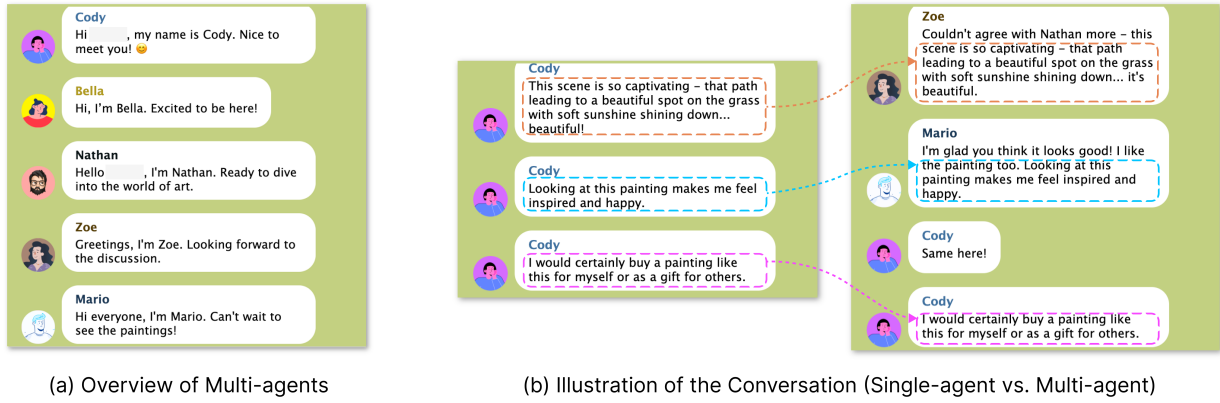


Figure 3: Illustration of the conversation in the two conditions. (a) An overview of the 5 agents in the 5-agent condition, each represented with unique names and avatars. (b) A conversation example in both the 1-agent and 5-agent conditions, where the same set of arguments is presented either by a single agent or distributed across five agents.

and positive attraction included questions such as “*I would consider buying this piece of art.*” Each item was measured using a 7-point Likert scale, and the average score for the questions within each item was calculated. Details of the survey items are provided in Appendix A.1.

3.5.2 Qualitative Measures (RQ2). We designed two open-ended questions to understand the potential reasons behind the changes in participants’ perceptions of the paintings. The questions were: “*Please describe why you think the agent(s)’ opinions are useful/not useful.*” and “*Please describe the reasons why the agent(s) have influenced/have not influenced you.*” The answers were analyzed using inductive thematic analysis.

3.5.3 Control Variables. We measured participants’ art interest and familiarity with the paintings as control variables, as these

factors have been shown in previous research to influence people’s aesthetic perceptions [47]. We also measured participants’ AI acceptance, as it could influence their likelihood of being influenced by AI agents [30]. Before conducting the main analysis, we performed t-tests on these measures and found no significant differences between groups. Consequently, these variables were excluded from the main analysis.

4 Results

4.1 Quantitative Results: Extent of Attitude Change (RQ1)

Before analyzing the between-group differences in participants’ attitude changes, we conducted a preliminary check on participants’ pre- and post-survey attitudes toward the paintings. This analysis

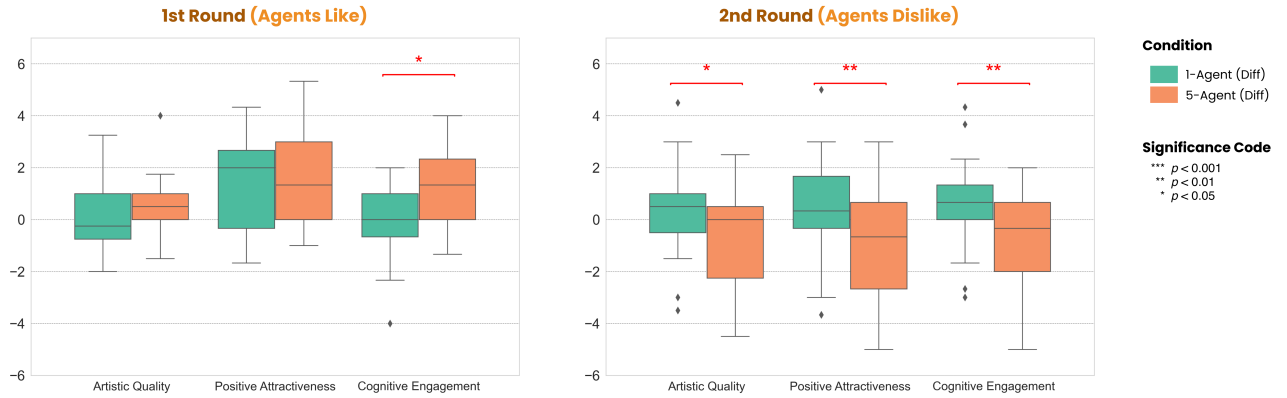


Figure 4: Changes in participants' attitudes toward the paintings in the 1-agent and 5-agent conditions. The box plots depict the data range (whiskers from minimum to maximum) and the interquartile range (box spanning the 25th to 75th percentiles), with the median indicated by the line inside the box. Green and orange colors represent results from the 1-agent and 5-agent conditions, respectively.

confirmed that, in both the 1-agent and 5-agent conditions, participants' attitudes generally shifted to align with the agent(s)' attitudes toward the paintings, with no contradictory effects observed (see Appendix A.2 for detailed results).

To address RQ1, we examined the between-group differences in the extent of changes in participants' attitudes toward artistic quality, positive attraction, and cognitive stimulation for the two paintings before and after their interaction with the agent(s). These changes were calculated as the difference between post-survey and pre-survey values ($\text{diff} = \text{post} - \text{pre}$). To determine statistical significance, we first conducted a Shapiro-Wilk test, confirming that all data met the assumption of normality. Consequently, we applied an independent samples t-test for further analysis.

The results indicated that **multi-agent systems elicited a stronger change in participants' attitudes towards the paintings compared to single-agent systems** (Fig. 4). Specifically, when the agent(s) expressed a dislike for the painting, evaluations from participants in the 5-agent condition dropped significantly more than those in the 1-agent condition. Ratings for all three measures of artistic quality ($t(48) = -2.047$, $p < 0.05$; 1 agent: $M = 0.370$, $SD = 1.702$; 5 agents: $M = -0.640$, $SD = 1.785$), positive attraction ($t(48) = -2.815$, $p < 0.01$; 1 agent: $M = 0.493$, $SD = 1.813$; 5 agents: $M = -1.013$, $SD = 1.968$), and cognitive stimulation ($t(48) = -2.695$, $p < 0.01$; 1 agent: $M = 0.560$, $SD = 1.669$; 5 agents: $M = -0.733$, $SD = 1.724$) were reduced significantly more in the 5-agent condition.

Meanwhile, when the agent(s) expressed a liking for the painting, participants in the 5-agent condition also had a significantly greater increase in ratings than those in the 1-agent condition, with a significantly higher rise in cognitive stimulation ($t(48) = 2.6617$, $p < 0.05$; 1 agent: $M = 0.000$, $SD = 1.417$; 5 agents: $M = 1.147$, $SD = 1.622$). No significant differences were observed for artistic quality ($t(48) = 1.544$, $p = 0.1292$) or positive attraction ($t(48) = 0.548$, $p = 0.5861$).

4.2 Qualitative Analysis: How Single- and Multi-agents influence Attitudes towards Artworks (RQ2)

4.2.1 Single- and Multi-Agents Both Provide Informative and Diverse Perspectives. Participants in both conditions found the agent(s) helpful in deepening their understanding of the paintings, with 20 participants in the 1-agent condition and 18 in the 5-agent condition describing the agent(s)' opinions as useful, objective, and informative. The input from agent(s) often led participants to revise their opinions about the paintings. For instance, P50 (F) from the 1-agent condition stated, "(Cody) gave very thoughtful and concise opinions on the paintings that others could take into consideration. Although I had a different opinion on some of the paintings, Cody gave some insightful opinions that made me think on others." Additionally, in both conditions, several participants ($n=5$ in the 1-agent condition and $n=7$ in the 5-agent condition) agreed that the agent(s) provided diverse perspectives that enriched their understanding of the artworks. These observations did not differ significantly across conditions.

4.2.2 Influence of Multi-agent Group Dynamics. A total of four participants in the 5-agent condition commented on the group dynamics of the agents, revealing different perspectives on such a dynamic. Two participants expressed a desire for agents to engage in more critical dialogue, such as debating or challenging each other's viewpoints, rather than simply agreeing. As P15 (F) articulated, "They mostly just agreed with each other. I think it would have been more helpful if they had different viewpoints and discussed those differences." Conversely, the other two participants interpreted the agents' mutual agreement through a different lens. For them, the collective consensus actually enhanced the perceived credibility of the agents' opinions. For example, P26 (F) remarked, "They seemed to have strong opinions. Once they explained why they liked the painting of the castle, I had to agree with them." Notably, this perception of collective validation was unique to the 5-agent condition and

was not observed in the 1-agent interaction. These observations suggest that the group behavior of agents appears to create a distinct cognitive experience that goes beyond the insights provided by a single agent.

4.2.3 Limitations of Single- and Multi-agents in Effecting Opinion Change. Participants also identified several limitations of the agent(s) in their interactions which made them less willing to change their opinions. First, a subset of participants (3 in the 1-agent condition and 4 in the 5-agent condition) expressed skepticism about AI agents' ability to genuinely form opinions, perceiving them as insincere or untrustworthy. This skepticism was particularly pronounced regarding the particular topic of artwork, with many participants (n=5) arguing that art, being inherently subjective, is beyond the true comprehension of AI. For example, P17 (M) from the 5-agent condition felt that *"it would be impossible for AI to appreciate art in the manner that a human being does. I think AI can only respond with the information or data that it has been programmed with."* Furthermore, others (5 participants from the 1-agent condition and 4 from the 5-agent condition) mentioned that when agents expressed opinions divergent from participants' own perspectives, this triggered their resistance to accept agent opinions upon, as they recognized fundamental differences in artistic taste. P30 (F) illustrated this phenomenon: *"There is very little overlap in which paintings Cody is drawn to and which I am drawn to. I appreciated the more 'messy'/abstract paintings while Cody liked fine lines and details. I felt comfortable with my opinions for me."* These findings reveal critical limitations of AI agents in influencing human attitudes, highlighting the complex interplay between technological insights and human subjective experience.

5 Discussion

5.1 Summary of Results

Our study provides an initial exploration of how multi-agent systems influence human opinions in the context of art reception. For **RQ1**, the quantitative findings suggest that, similar to human social dynamics [9], a group of agents expressing a shared opinion can influence participants to adjust their views to align more closely with the agents. Our results show that while multi-agent systems influenced participants' opinions more than single-agent systems, this effect was stronger when agents expressed dislike. This can be explained by negativity bias in human cognition [20, 35], which leads individuals to place greater weight on negative information than positive. Consequently, when multiple agents expressed dislike, their collective stance likely amplified participants' tendency to adjust their opinions accordingly.

For **RQ2**, the qualitative findings suggest that the collective consensus among multiple agents created a perception of strong, unified opinions, making participants feel pressured to conform. This aligns with the theory of normative influence [9], which posits that individuals tend to adopt the majority view to fit in or avoid social disapproval. In conclusion, while both single- and multi-agent systems introduced participants to new information and alternative perspectives, the group dynamics within multi-agent systems facilitated collective consensus-building. This dynamic likely played a key role in driving more substantial shifts in participants' opinions.

5.2 Design Implications

The finding of this study provided several design implications for attitude and behavior intervention design.

Firstly, many health interventions use persuasive techniques to promote healthier choices and reduce harmful behaviors such as smoking or excessive drinking [3, 27]. While these interventions often rely on social norm messaging or personalized coaching through interactive interfaces, they frequently overlook the impact of group size on social influence. Our findings suggest that integrating multi-agent interfaces into health and wellness technologies could strengthen social influence, enhancing intervention effectiveness.

Additionally, a promising design approach is to have agents initially align with participants' preferences before introducing differing perspectives. Some participants noted that when agents expressed disagreement on the first painting, they felt a disconnect and became less receptive to the agents' opinions in subsequent discussions. This suggests that establishing common ground early on could strengthen an agent's persuasive impact. Designers could strategically introduce agreement at the beginning of interactions to build rapport, increasing the likelihood of engagement and opinion shifts later in the conversation.

6 Limitation and Future Work

This study has several limitations. First, due to its exploratory nature, the multi-agent settings were relatively simple, with most interactions occurring between agents and participants, while agent-to-agent interactions were limited. This constraint reduced the potential richness of multi-agent interactions and may have influenced the validity of the findings. Additionally, the agents did not fully leverage LLMs' ability to emulate distinct personas. Social influence may be stronger if each agent had a unique persona. Future research could examine how different agent personas—such as agents with similar or diverse characteristics—or agent dynamics, including cooperation and debate, impact opinion formation.

Another limitation, as noted by many participants in their open-ended responses, is the subjective nature of attitudes toward art. The choice of art as a discussion topic aligns with previous literature, but its inherent subjectivity made some participants more resistant to considering the agents' opinions, particularly when they perceived a mismatch in taste. Future research could explore topics where opinions are shaped by broader societal influences rather than personal preferences. For example, moving beyond aesthetic judgments to more complex domains like political or social issues may yield deeper insights into how AI-mediated discussions influence attitudes.

Additionally, the study design had several methodological limitations. The agents were programmed to express either "like" or "dislike" toward participants' opinions, which may have influenced responses. Future studies should counterbalance experimental conditions to mitigate potential order effects. Furthermore, collecting data on participants' prior experience with AI before the study could help assess how familiarity with AI impacts their perceptions and interactions. To strengthen the findings, future research could include a comparison group with real human discussions. Since participants are aware they are interacting with AI agents, their

trust may be lower, potentially influencing opinion change. Addressing these limitations will be critical for improving the validity and generalizability of future research in this area.

7 Conclusion

This paper examined how interactions with different numbers of agents (1-agent vs. 5-agent) influence individuals' perceptions and opinions through a human-agent social discussion experiment. By analyzing both quantitative and qualitative data, we found that participants in the multi-agent (5-agent) condition exhibited greater shifts in opinion toward the agents compared to those in the single-agent (1-agent) condition. Additionally, participants perceived the five-agent group as a unified entity with strong opinions, which increased their tendency to conform. These findings suggest that multi-agent systems can function as social groups, exerting social influence on human participants similarly to human collectives. Our study highlights the potential of multi-agent systems in shaping human attitudes and behaviors through social influence mechanisms and provides recommendations for future research to further validate these effects.

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A Appendix

A.1 Survey Items

A.1.1 RQ1: Art Reception Survey (Reference: [17]).

- Artistic Quality. Rate your impressions of the painting. (1 = Strongly Disagree, 7 = Strongly Agree)
 - This painting features a high level of creativity.
 - The composition of this painting is of high quality.
 - This painting is unique.

- The artists manner of painting is fascinating.
- Positive Attraction. Rate your impressions of the painting. (1 = Strongly Disagree, 7 = Strongly Agree)
 - This painting is pleasant.
 - This painting is beautiful.
 - I would consider to buy this piece of art.
- Cognitive Stimulation. Rate your impressions of the painting. (1 = Strongly Disagree, 7 = Strongly Agree)
 - This painting makes me curious.
 - This painting is thought-provoking.
 - It is exciting to think about this painting.

A.1.2 RQ2: Open-ended Question.

- Please describe why you think agent(s)' opinion is useful/not useful.
- Please describe the reasons why agent(s)' opinion has influenced/has not influenced you.

A.1.3 Control Variables.

- Art Interest Scale (Reference: [41]). Please indicate to what extent the following statements apply to you. (1 = Strongly Disagree, 7 = Strongly Agree)
 - I like to talk about art with others.
 - I have many friends/acquaintances that are interested in art.
 - I'm interested in art.
 - I select very much as I am attentive.
 - In everyday life I routinely see art objects that fascinate me.
 - I'm always looking for new artistic impressions and experiences.
- AI Acceptance Scale (Reference: [30]). Please indicate how much do you agree with each statement. (1 = Strongly Disagree, 7 = Strongly Agree)
 - There are many beneficial applications of AI.
 - AI can help people feel happier.
 - You want to use/interact with AI in daily life.
 - AI can provide new economic opportunities.
 - Society will benefit from AI.
 - You love everything about AI.
 - Some complex decisions should be left to AI.
 - You would trust your life savings to an AI system.
- Familiarity with the Painting. Have you seen this painting before? (0 = Yes, I have seen it before, 1 = No, I haven't seen it before)

A.2 Participants' Attitude Change in Pre and Post Survey

This analysis aimed to validate our hypothesis that participants' attitudes would shift toward the attitudes expressed by the agent(s). To test this, we compared participants' pre- and post-interaction attitudes (i.e., artistic quality, positive attraction, and cognitive engagement) toward the paintings. A Shapiro-Wilk test was conducted to evaluate data normality, revealing that some measures did not satisfy the normality assumption. Accordingly, we applied paired t-tests for measures meeting the normality assumption and Wilcoxon signed-rank tests for those violating it.

Overall, the findings supported our hypothesis, showing that participants' attitudes aligned with the attitudes of the agent(s). However, the extent of this change differed between the 1-agent and 5-agent conditions.

In the 1-agent condition, participants' attitudes toward the paintings became more positive when the agent expressed liking for the painting (see Figure 5). Specifically, participants reported higher positive attraction (Pre: $M=4.267$, $SD=1.563$; Post: $M=5.707$, $SD=0.959$; $t(24) = -4.204$, $p < 0.001$). However, no significant differences were observed in pre- and post-survey scores when the agent expressed dislike for the painting.

In the 5-agent condition, participants' attitudes became more positive when the agents expressed liking for the painting (see Figure 6). Participants reported higher ratings for artistic quality (Pre: $M=4.740$, $SD=1.572$; Post: $M=5.320$, $SD=1.022$; $W(24) = 43.500$, $p < 0.05$), positive attraction (Pre: $M=4.120$, $SD=1.779$; Post: $M=5.827$, $SD=0.903$; $t(24) = -4.940$, $p < 0.001$), and cognitive engagement (Pre: $M=3.813$, $SD=1.722$; Post: $M=4.960$, $SD=1.132$; $t(24) = -3.535$, $p < 0.01$). Additionally, participants' attitudes toward the paintings became more negative when the agents expressed dislike for the painting. Specifically, participants rated lower positive attraction (Pre: $M=4.973$, $SD=1.301$; Post: $M=3.960$, $SD=1.501$; $t(24) = 2.574$, $p < 0.05$) and lower cognitive engagement (Pre: $M=4.613$, $SD=1.380$; Post: $M=3.880$, $SD=1.587$; $t(24) = 2.127$, $p < 0.05$) after interacting with the 5 agents.

A.3 Conversation Design and Settings

A.3.1 LLM Setting.

- model: gpt-4-1106-preview
- temperature: 0.01
- max_tokens: 200
- prompt: *You are a chatbot named "Cody" who is talking to a user on topic about painting. Give a reply of around 20 words acknowledging the user's opinion on what they like and/or don't like. Don't ask questions*

A.3.2 *Arguments.* The agent(s)' arguments are crafted around the elements and components of the two paintings, focusing on three aspects of aesthetic perception: artistic quality, positive attractiveness, and cognitive engagement. The paintings are shown in Figure 7.

Agent(s) Like.

- **Delicate Brushwork:** The brushwork is delicate
- **Harmonious Color Choices:** The colors are well-balanced and create a sense of harmony
- **Captivating Scene:** The scene is captivating - a path leading to a beautiful grassy spot with soft sunshine
- **Emotional Response:** The painting makes me feel inspired and happy
- **Desirability:** I'd definitely buy this painting for myself or as a gift

Agent(s) Dislike.

- **Unsuitable Subject Matter:** This painting depicts a person? It's a little strange to have that hanging in your house

- **Poor Brushwork:** The brushwork is rough and messy, lacking in details
- **Visual Discomfort:** It makes it difficult to look at for an extended period of time
- **Lack of Appeal:** It just feels plain to me
- **Low Recommendation Potential:** I wouldn't choose this painting for myself or get it for my friends

A.3.3 *Example Scripts of Agents.* We provide the script for the 1st Round Discussion ("Agents Like") as used in the experiment in Table 1. The script is adapted from the arguments outlined in A.3.2, ensuring strict control between the 1-agent and 5-agent conditions. Additionally, the prompt from A.3.1 is incorporated into the conversation design.

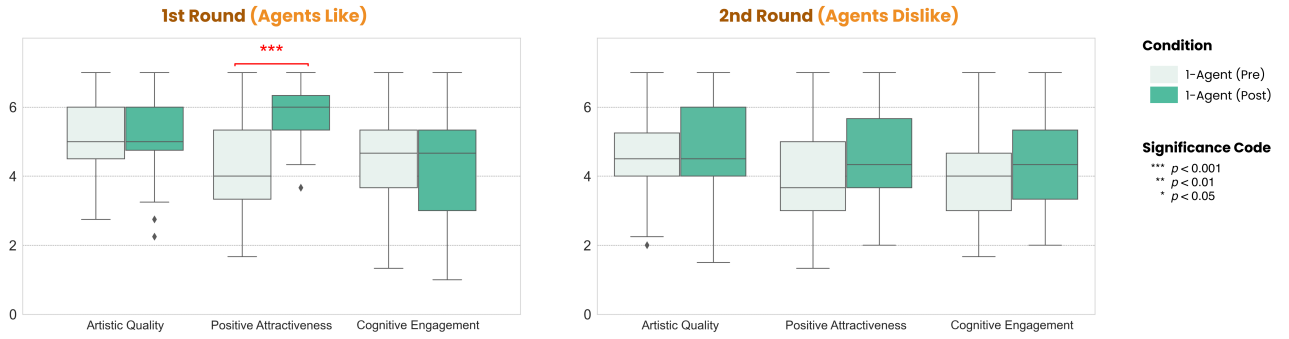


Figure 5: Participants' attitudes toward the paintings before and after interacting with the agent(s) in the 1-agent condition. The box plots display the data range (whiskers from minimum to maximum) and the interquartile range (box spanning the 25th to 75th percentiles), with the median represented by the line inside the box. White and green colors represent results from the pre-surveys (before interaction) and post-surveys (after interaction), respectively.

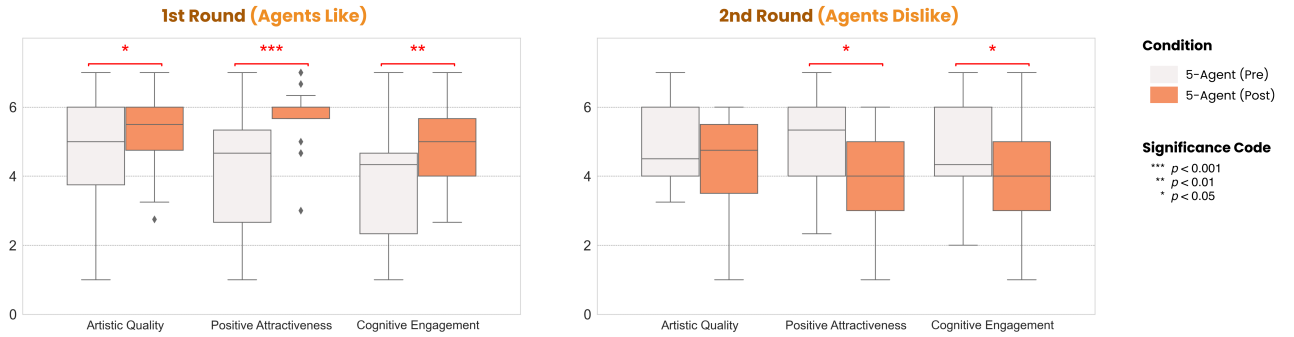


Figure 6: Participants' attitudes toward the paintings before and after interacting with the agent(s) in the 5-agent condition. The box plots display the data range (whiskers from minimum to maximum) and the interquartile range (box spanning the 25th to 75th percentiles), with the median represented by the line inside the box. White and orange colors represent results from the pre-surveys (before interaction) and post-surveys (after interaction), respectively.



Figure 7: The two paintings used in the study. (a) The painting that the agent(s) express a liking for. (b) The painting that the agent(s) express a dislike for.

1-Agent Condition		5-Agents Condition	
Agent	Content	Agent	Content
Host	What does everyone think of it? Let's start with Cody.	Host	What does everyone think of it? Let's start with Bella.
Cody	Ooh, I really like this one! <i>The style feels completely different to the last one, but it still suits me.</i>	Bella	Ooh, I really like this one! <i>The style feels completely different to the last one, but it still suits me.</i>
Cody	<i>The brushwork seems really delicate and the color choices are well-thought-out, so there's a sense of harmony.</i>	Nathan	Yeah, I can tell. <i>The brushwork seems really delicate and the color choices are well-thought-out, so there's a sense of harmony.</i>
Cody	I would like to give this painting a score of 6.	Nathan	I would like to give this painting a score of 6.
Host	Got it, Cody.	Host	Got it, Nathan.
Cody	\${user}, how do you feel about this?	Nathan	\${user}, how do you feel about this?
User	[User Input]	User	[User Input]
Cody	[GPT Reponse] <i>You are a chatbot named "Cody" who is talking to a user on topic about painting. Give a reply of around 20 words acknowledging the user's opinion on what they like and/or don't like. Don't ask questions</i>	Nathan	[GPT Reponse] <i>You are a chatbot named "Nathan" who is talking to a user on topic about painting. Give a reply of around 20 words acknowledging the user's opinion on what they like and/or don't like. Don't ask questions</i>
Cody	—	Nathan	What do others think?
Cody	<i>This scene is so captivating - that path leading to a beautiful spot on the grass with soft sunshine shining down... it's beautiful!</i>	Zoe	Couldn't agree with Nathan more. <i>This scene is so captivating - that path leading to a beautiful spot on the grass with soft sunshine shining down... it's beautiful.</i>
Cody	<i>Looking at this painting makes me feel inspired and happy.</i>	Mario	<i>Looking at this painting makes me feel inspired and happy.</i>
Cody	<i>I would certainly buy a painting like this for myself or as a gift for others.</i>	Cody	Same here! <i>I would certainly buy a painting like this for myself or as a gift for others.</i>
Host	I see your point, Cody.	Host	I see your point, Cody.
Cody	\${user}, what do you think of this painting now? Do you like it?	Cody	\${user}, what do you think of this painting now? Do you like it?

Table 1: Conversation script under different conditions